

Retail Customer Intelligence Dashboard

Project Report — Python, SQL and Power BI Analysis

1. Overview

This report analyzes 3,900 retail transactions to understand what drives customer spending and how the business can act on it. The work spans three stages: cleaning and preparing the data in Python, answering structured business questions in SQL, and presenting the results through an interactive Power BI dashboard.

2. Dataset Summary

3,900 rows, 18 columns, covering customer demographics, item and category purchased, purchase amount, location, season, review rating, subscription status, shipping type, discount usage, payment method, and purchase frequency. The Review Rating column had 37 missing values, imputed using the median rating for each product category.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	39
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	1
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Summary statistics from df.describe(include='all') — part 1

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Summary statistics — discount, promo, purchase frequency fields

3. Data Preparation (Python)

- Loaded and inspected the dataset with pandas, using df.info() and df.describe() to confirm structure and check for anomalies.

- Handled missing review ratings by imputing the category-level median rather than the global mean, to avoid skewing lower-priced or higher-priced categories.
- Standardized column names to snake_case for consistency across the Python and SQL stages.
- Engineered an age_group column (Young Adult, Adult, Middle-aged, Senior) and a purchase_frequency_days column derived from the stated purchase frequency.
- Checked for redundancy between Discount Applied and Promo Code Used — found them near-identical — and dropped the duplicate column to simplify downstream analysis.
- Loaded the cleaned dataset into PostgreSQL for structured querying.

4. Business Analysis (SQL)

Structured queries in PostgreSQL were used to answer specific business questions using joins, aggregation, CTEs, window functions, and ranking.

4.1 Revenue by gender

	gender text	revenue numeric
1	Female	75191
2	Male	157890

4.2 High-spending discount users

Customers who used a discount but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:00.133

4.3 Top 5 products by rating

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4.4 Shipping type comparison

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

4.5 Subscribers vs. non-subscribers

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Subscribers spend marginally less per order than non-subscribers, suggesting subscription status alone is not a strong revenue predictor — value likely comes from purchase frequency instead.

4.6 Customer revenue concentration (original analysis)

To understand how concentrated revenue is among top spenders, customers were ranked by total spend and checked against the classic 80/20 rule. The top 20% of customers by spend account for approximately 31% of total revenue — meaning revenue here is considerably more evenly distributed across the customer base than a typical Pareto pattern. This implies retention strategy should focus broadly rather than narrowly on a small VIP segment.

4.7 Discount-dependent products

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

4.8 Customer segmentation

Customers classified into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

4.9 Top 3 products per category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

4.10 Repeat buyers & subscriptions

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

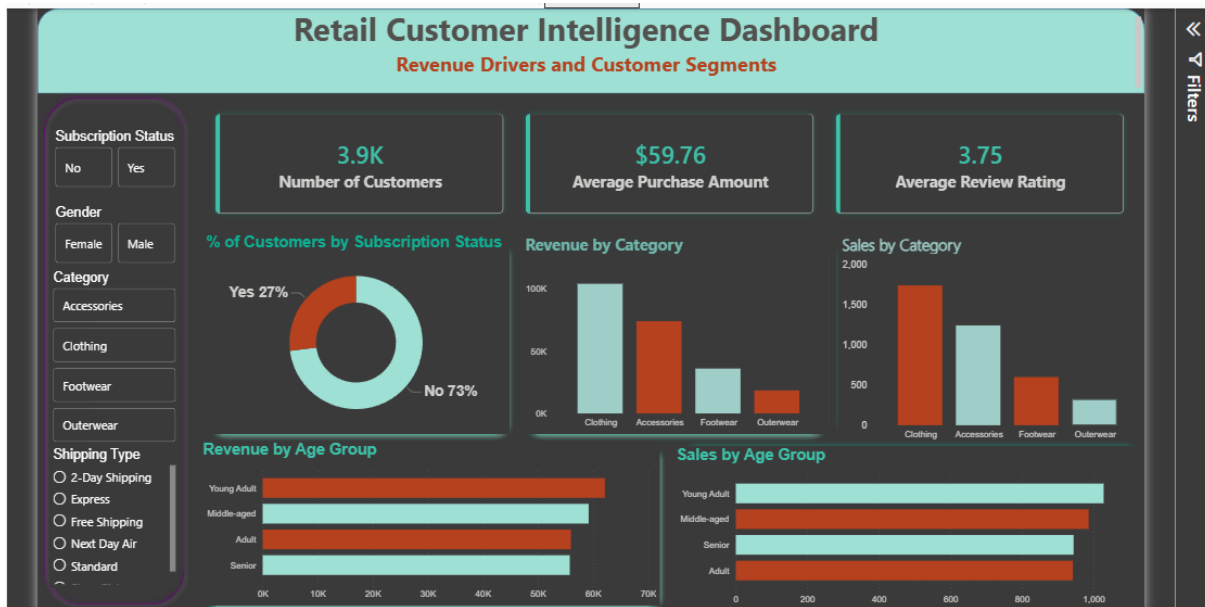
4.11 Revenue by age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Revenue is fairly evenly spread across age groups, with Young Adults contributing slightly more — a narrower gap than typically assumed in retail targeting.

5. Dashboard (Power BI)

An interactive dashboard was built in Power BI summarizing customer count, average order value, and average rating alongside revenue breakdowns by category and age group, with slicers for subscription status, gender, category, and shipping type to let stakeholders explore the data themselves.



Retail Customer Intelligence Dashboard — Power BI

6. Business Recommendations

- **Broaden retention efforts** — since revenue is not heavily concentrated in a small top-spender segment, loyalty programs should target the broader Loyal and Returning segments rather than a narrow VIP tier.
- **Revisit discount strategy** on Hat, Sneakers, Coat, Sweater, and Pants — high discount dependency on these items may be compressing margin unnecessarily.
- **Re-evaluate subscription incentives** — since subscribers do not out-spend non-subscribers per order, the subscription program should be positioned around frequency and retention rather than order value.
- **Balance marketing spend across age groups** rather than concentrating on a single demographic, given the relatively even revenue split.
- **Promote top-rated products** (Gloves, Sandals, Boots, Hat, Skirt) in campaigns to leverage existing customer trust.